

Day 3 — Subtract from a Power of 10

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute $1000-567$ and similar instantly.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilaṁ Navataścaramaṁ Daśataḥ* (IAST: Nikhilaṁ Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The clean application.** Whenever we’re subtracting from 10^k (10, 100, 1000, 10000, ...), Nikhilam gives the answer directly. No borrowing, no carries.
2. **Speed examples.**
 - $10 - 4 = 6$ (4 from 10)
 - $100 - 37 = 63$ (3 from 9, 7 from 10)
 - $1000 - 268 = 732$
 - $10000 - 1547 = 8453$
3. **Practical use case.** Computing change at a shop. “*You paid ₹1000 for an item costing ₹427. How much change?*” Nikhilam: $9-4=5$, $9-2=7$, $10-7=3 \rightarrow ₹573$. Mental math, no paper.
4. **Verify with traditional subtraction.** Have students cross-check 2 problems by the borrowing method. Same answer. The Nikhilam is just a smarter procedure for the same arithmetic.
5. **Class discussion.** “*When IS Nikhilam genuinely faster than borrowing? When does borrowing keep up?*” (Borrowing keeps up for small numbers like $10-4$. Nikhilam dominates for larger powers.)

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Subtract from a Power of 10.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).